US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12409033
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Zeng; Qinggang

Axisymmetric adjustable device for treating mitral regurgitation

Abstract

A prosthetic device for treating a native valve of a heart includes a sealing element and an anchoring element. The sealing element comprises a braided mesh material. The sealing element is dimensioned to be deployed at the native valve. The sealing element is configured to both be radially expanded and radially reduced while at a position between the native valve leaflets. The anchoring element is coupled to the sealing element and is configured to support the sealing element between the native valve leaflets.

Inventors: **Zeng; Qinggang (Mission Viejo, CA)**

Applicant: **Edwards Lifesciences Corporation** (Irvine, CA)

Family ID: **65630197**

Assignee: **EDWARDS LIFESCIENCES CORPORATION (Irvine, CA)**

Appl. No.: **17/378521**

Filed: **July 16, 2021**

Prior Publication Data

Document Identifier	Publication Date
US 20210338427 A1	Nov. 04, 2021

Related U.S. Application Data

continuation parent-doc US 16121507 20180904 US 11065117 child-doc US 17378521
us-provisional-application US 62555863 20170908

Publication Classification

Int. Cl.: **A61F2/24** (20060101)

U.S. Cl.:

CPC **A61F2/2445** (20130101); **A61F2/2454** (20130101); **A61F2/246** (20130101); **A61F2/2466** (20130101); A61F2210/0014 (20130101); A61F2220/0008 (20130101); A61F2220/0025 (20130101); A61F2230/0008 (20130101); A61F2230/0069 (20130101); A61F2250/0098 (20130101)

Field of Classification Search

CPC: A61F (2/2445); A61F (2/2454); A61F (2/246); A61F (2/2466); A61F (2210/0014); A61F (2220/0008); A61F (2220/0025); A61F (2230/0008); A61F (2230/0069); A61F (2250/0098)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3874388	12/1974	King et al.	N/A	N/A
4340091	12/1981	Skelton et al.	N/A	N/A
4506669	12/1984	Blake, III	N/A	N/A
4590937	12/1985	Deniega	N/A	N/A
4693248	12/1986	Failla	N/A	N/A
4803983	12/1988	Siegel	N/A	N/A
5125895	12/1991	Buchbinder et al.	N/A	N/A
5171252	12/1991	Friedland	N/A	N/A
5195962	12/1992	Martin et al.	N/A	N/A
5292326	12/1993	Green et al.	N/A	N/A
5327905	12/1993	Avitall	N/A	N/A
5363861	12/1993	Edwards et al.	N/A	N/A
5370685	12/1993	Stevens	N/A	N/A
5389077	12/1994	Melinyshyn et al.	N/A	N/A
5411552	12/1994	Andersen et al.	N/A	N/A
5450860	12/1994	O'Connor	N/A	N/A
5456674	12/1994	Bos et al.	N/A	N/A
5474057	12/1994	Makower et al.	N/A	N/A
5478353	12/1994	Yoon	N/A	N/A
5487746	12/1995	Yu et al.	N/A	N/A
5565004	12/1995	Christoudias	N/A	N/A
5607462	12/1996	Imran	N/A	N/A
5609598	12/1996	Laufer et al.	N/A	N/A
5611794	12/1996	Sauer et al.	N/A	N/A
5626607	12/1996	Malecki et al.	N/A	N/A
5695504	12/1996	Gifford, III et al.	N/A	N/A
5716417	12/1997	Girard et al.	N/A	N/A
5727569	12/1997	Benetti et al.	N/A	N/A
5741297	12/1997	Simon	N/A	N/A
5782746	12/1997	Wright	N/A	N/A
5797960	12/1997	Stevens et al.	N/A	N/A
5836311	12/1997	Borst et al.	N/A	N/A

5843076	12/1997	Webster, Jr. et al.	N/A	N/A
5855590	12/1998	Malecki et al.	N/A	N/A
5885271	12/1998	Hamilton et al.	N/A	N/A
5888247	12/1998	Benetti	N/A	N/A
5891017	12/1998	Swindle et al.	N/A	N/A
5891112	12/1998	Samson	N/A	N/A
5894843	12/1998	Benetti et al.	N/A	N/A
5921979	12/1998	Kovac et al.	N/A	N/A
5944738	12/1998	Amplatz et al.	N/A	N/A
5957835	12/1998	Anderson et al.	N/A	N/A
5972020	12/1998	Carpentier et al.	N/A	N/A
5980534	12/1998	Gimpelson	N/A	N/A
6004329	12/1998	Myers et al.	N/A	N/A
6010531	12/1999	Donlon et al.	N/A	N/A
6017358	12/1999	Yoon et al.	N/A	N/A
6086600	12/1999	Kortenbach	N/A	N/A
6120496	12/1999	Whayne et al.	N/A	N/A
6132370	12/1999	Furnish et al.	N/A	N/A
6162239	12/1999	Manhes	N/A	N/A
6165183	12/1999	Kuehn et al.	N/A	N/A
6182664	12/2000	Cosgrove	N/A	N/A
6193732	12/2000	Frantzen et al.	N/A	N/A
6193734	12/2000	Bolduc et al.	N/A	N/A
6200315	12/2000	Gaiser et al.	N/A	N/A
6228032	12/2000	Eaton et al.	N/A	N/A
6241743	12/2000	Levin et al.	N/A	N/A
6269819	12/2000	Oz et al.	N/A	N/A
6269829	12/2000	Chen et al.	N/A	N/A
6312447	12/2000	Grimes	N/A	N/A
6461366	12/2001	Seguin	N/A	N/A
6468285	12/2001	Hsu et al.	N/A	N/A
6508806	12/2002	Hoste	N/A	N/A
6508825	12/2002	Selmon et al.	N/A	N/A
6530933	12/2002	Yeung et al.	N/A	N/A
6537290	12/2002	Adams et al.	N/A	N/A
6544215	12/2002	Bencini et al.	N/A	N/A
6626930	12/2002	Allen et al.	N/A	N/A
6629534	12/2002	St. Goar et al.	N/A	N/A
6719767	12/2003	Kimblad	N/A	N/A
6764510	12/2003	Vidlund et al.	N/A	N/A
6770083	12/2003	Seguin	N/A	N/A
6837867	12/2004	Kortelling	N/A	N/A
6855137	12/2004	Bon	N/A	N/A
6913614	12/2004	Marino et al.	N/A	N/A
6939337	12/2004	Parker et al.	N/A	N/A
6945956	12/2004	Waldhauser et al.	N/A	N/A
7048754	12/2005	Martin et al.	N/A	N/A
7101395	12/2005	Tremulis et al.	N/A	N/A
7125421	12/2005	Tremulis et al.	N/A	N/A
7288097	12/2006	Seguin	N/A	N/A

7371210	12/2007	Brock et al.	N/A	N/A
7464712	12/2007	Oz et al.	N/A	N/A
7509959	12/2008	Oz et al.	N/A	N/A
7569062	12/2008	Kuehn et al.	N/A	N/A
7618449	12/2008	Tremulis et al.	N/A	N/A
7682369	12/2009	Seguin	N/A	N/A
7731706	12/2009	Potter	N/A	N/A
7744609	12/2009	Allen et al.	N/A	N/A
7748389	12/2009	Salahieh et al.	N/A	N/A
7753932	12/2009	Gingrich et al.	N/A	N/A
7758596	12/2009	Oz et al.	N/A	N/A
7780723	12/2009	Taylor	N/A	N/A
7803185	12/2009	Gabbay	N/A	N/A
7824443	12/2009	Salahieh et al.	N/A	N/A
7981123	12/2010	Seguin	N/A	N/A
7988724	12/2010	Salahieh et al.	N/A	N/A
8052750	12/2010	Tuval et al.	N/A	N/A
8070805	12/2010	Vidlund et al.	N/A	N/A
8096985	12/2011	Legaspi et al.	N/A	N/A
8104149	12/2011	McGarity	N/A	N/A
8133239	12/2011	Oz et al.	N/A	N/A
8147542	12/2011	Maisano et al.	N/A	N/A
8172856	12/2011	Eigler et al.	N/A	N/A
8206437	12/2011	Bonhoeffer et al.	N/A	N/A
8216301	12/2011	Bonhoeffer et al.	N/A	N/A
8303653	12/2011	Bonhoeffer et al.	N/A	N/A
8313525	12/2011	Tuval et al.	N/A	N/A
8348995	12/2012	Tuval et al.	N/A	N/A
8348996	12/2012	Tuval et al.	N/A	N/A
8414643	12/2012	Tuval et al.	N/A	N/A
8425404	12/2012	Wilson et al.	N/A	N/A
8449599	12/2012	Chau et al.	N/A	N/A
8449606	12/2012	Eliasen et al.	N/A	N/A
8460368	12/2012	Taylor et al.	N/A	N/A
8470028	12/2012	Thornton et al.	N/A	N/A
8480730	12/2012	Maurer et al.	N/A	N/A
8540767	12/2012	Zhang	N/A	N/A
8579965	12/2012	Bonhoeffer et al.	N/A	N/A
8585756	12/2012	Bonhoeffer et al.	N/A	N/A
8652202	12/2013	Alon et al.	N/A	N/A
8668733	12/2013	Haug et al.	N/A	N/A
8721665	12/2013	Oz et al.	N/A	N/A
8740918	12/2013	Seguin	N/A	N/A
8771347	12/2013	DeBoer et al.	N/A	N/A
8778017	12/2013	Eliasen et al.	N/A	N/A
8834564	12/2013	Tuval et al.	N/A	N/A
8840663	12/2013	Salahieh et al.	N/A	N/A
8876894	12/2013	Tuval et al.	N/A	N/A
8876895	12/2013	Tuval et al.	N/A	N/A
8945177	12/2014	Dell et al.	N/A	N/A

8968395	12/2014	Hauser et al.	N/A	N/A
9034032	12/2014	McLean et al.	N/A	N/A
9198757	12/2014	Schroeder et al.	N/A	N/A
9220507	12/2014	Patel et al.	N/A	N/A
9259317	12/2015	Wilson et al.	N/A	N/A
9282972	12/2015	Patel et al.	N/A	N/A
9301834	12/2015	Tuval et al.	N/A	N/A
9308360	12/2015	Bishop et al.	N/A	N/A
9387071	12/2015	Tuval et al.	N/A	N/A
9414918	12/2015	Chau et al.	N/A	N/A
9427327	12/2015	Parrish	N/A	N/A
9439763	12/2015	Geist et al.	N/A	N/A
9510837	12/2015	Seguin	N/A	N/A
9510946	12/2015	Chau et al.	N/A	N/A
9572660	12/2016	Braido et al.	N/A	N/A
9610163	12/2016	Khairkhahan et al.	N/A	N/A
9642704	12/2016	Tuval et al.	N/A	N/A
9700445	12/2016	Martin et al.	N/A	N/A
9775963	12/2016	Miller	N/A	N/A
D809139	12/2017	Marsot et al.	N/A	N/A
9889002	12/2017	Bonhoeffer et al.	N/A	N/A
9949824	12/2017	Bonhoeffer et al.	N/A	N/A
10076327	12/2017	Ellis et al.	N/A	N/A
10076415	12/2017	Metchik et al.	N/A	N/A
10099050	12/2017	Chen et al.	N/A	N/A
10105221	12/2017	Siegel	N/A	N/A
10105222	12/2017	Metchik et al.	N/A	N/A
10111751	12/2017	Metchik et al.	N/A	N/A
10123873	12/2017	Metchik et al.	N/A	N/A
10130475	12/2017	Metchik et al.	N/A	N/A
10136993	12/2017	Metchik et al.	N/A	N/A
10159570	12/2017	Metchik et al.	N/A	N/A
10226309	12/2018	Ho et al.	N/A	N/A
10231837	12/2018	Metchik et al.	N/A	N/A
10238493	12/2018	Metchik et al.	N/A	N/A
10238494	12/2018	McNiven et al.	N/A	N/A
10238495	12/2018	Marsot et al.	N/A	N/A
10299924	12/2018	Kizuka	N/A	N/A
10376673	12/2018	Van Hoven et al.	N/A	N/A
10575841	12/2019	Paulos	N/A	N/A
2001/0005787	12/2000	Oz et al.	N/A	N/A
2002/0013571	12/2001	Goldfarb et al.	N/A	N/A
2002/0107531	12/2001	Schreck et al.	N/A	N/A
2002/0173811	12/2001	Tu et al.	N/A	N/A
2002/0183787	12/2001	Wahr et al.	N/A	N/A
2003/0144573	12/2002	Heilman et al.	N/A	N/A
2003/0187467	12/2002	Schreck	N/A	N/A
2003/0208231	12/2002	Williamson et al.	N/A	N/A
2004/0003819	12/2003	St. Goar et al.	N/A	N/A
2004/0030382	12/2003	St. Goar et al.	N/A	N/A

2004/0034365	12/2003	Lentz et al.	N/A	N/A
2004/0044350	12/2003	Martin et al.	N/A	N/A
2004/0044365	12/2003	Bachman	N/A	N/A
2004/0049207	12/2003	Goldfarb et al.	N/A	N/A
2004/0122448	12/2003	Levine	N/A	N/A
2004/0127981	12/2003	Rahdert et al.	N/A	N/A
2004/0127982	12/2003	Machold et al.	N/A	N/A
2004/0147943	12/2003	Kobayashi	N/A	N/A
2004/0181135	12/2003	Drysen	N/A	N/A
2004/0181206	12/2003	Chiu et al.	N/A	N/A
2004/0181238	12/2003	Zarbatany et al.	N/A	N/A
2004/0210307	12/2003	Khairkhahan	N/A	N/A
2004/0220593	12/2003	Greenhalgh	N/A	N/A
2005/0010287	12/2004	Macoviak et al.	N/A	N/A
2005/0049618	12/2004	Masuda et al.	N/A	N/A
2005/0070926	12/2004	Ortiz	N/A	N/A
2005/0137690	12/2004	Salahieh et al.	N/A	N/A
2005/0143767	12/2004	Kimura et al.	N/A	N/A
2005/0165429	12/2004	Douglas et al.	N/A	N/A
2005/0216039	12/2004	Lederman	N/A	N/A
2005/0251177	12/2004	Saadat et al.	N/A	N/A
2005/0251183	12/2004	Buckman et al.	N/A	N/A
2005/0288786	12/2004	Chanduszko	N/A	N/A
2006/0020275	12/2005	Goldfarb et al.	N/A	N/A
2006/0058871	12/2005	Zakay et al.	N/A	N/A
2006/0089671	12/2005	Goldfarb et al.	N/A	N/A
2006/0100649	12/2005	Hart	N/A	N/A
2006/0122647	12/2005	Callaghan et al.	N/A	N/A
2006/0142694	12/2005	Bednarek et al.	N/A	N/A
2006/0173251	12/2005	Govari et al.	N/A	N/A
2006/0178700	12/2005	Quinn	N/A	N/A
2006/0224169	12/2005	Weisenburgh et al.	N/A	N/A
2007/0010800	12/2006	Weitzner et al.	N/A	N/A
2007/0010877	12/2006	Salahieh et al.	N/A	N/A
2007/0016286	12/2006	Herrmann et al.	N/A	N/A
2007/0021779	12/2006	Garvin et al.	N/A	N/A
2007/0032807	12/2006	Ortiz et al.	N/A	N/A
2007/0093857	12/2006	Rogers et al.	N/A	N/A
2007/0093890	12/2006	Eliasen et al.	N/A	N/A
2007/0156197	12/2006	Root et al.	N/A	N/A
2007/0191154	12/2006	Genereux et al.	N/A	N/A
2007/0197858	12/2006	Goldfarb et al.	N/A	N/A
2007/0198038	12/2006	Cohen et al.	N/A	N/A
2007/0265700	12/2006	Eliasen et al.	N/A	N/A
2007/0282414	12/2006	Soltis et al.	N/A	N/A
2007/0282429	12/2006	Hauser et al.	N/A	N/A
2007/0293943	12/2006	Quinn	N/A	N/A
2007/0299387	12/2006	Williams et al.	N/A	N/A
2007/0299424	12/2006	Cumming et al.	N/A	N/A
2008/0039743	12/2007	Fox et al.	N/A	N/A

2008/0039953	12/2007	Davis et al.	N/A	N/A
2008/0065149	12/2007	Thielen et al.	N/A	N/A
2008/0077144	12/2007	Crofford	N/A	N/A
2008/0091169	12/2007	Heideman et al.	N/A	N/A
2008/0140089	12/2007	Kogiso et al.	N/A	N/A
2008/0147093	12/2007	Roskopf et al.	N/A	N/A
2008/0147112	12/2007	Sheets et al.	N/A	N/A
2008/0149685	12/2007	Smith et al.	N/A	N/A
2008/0167713	12/2007	Bolling	N/A	N/A
2008/0177300	12/2007	Mas et al.	N/A	N/A
2008/0208332	12/2007	Lamphere et al.	N/A	N/A
2008/0221672	12/2007	Lamphere et al.	N/A	N/A
2008/0255427	12/2007	Satake et al.	N/A	N/A
2008/0281411	12/2007	Berreklouw	N/A	N/A
2008/0287862	12/2007	Weitzner et al.	N/A	N/A
2008/0294247	12/2007	Yang et al.	N/A	N/A
2008/0312506	12/2007	Spivey et al.	N/A	N/A
2008/0319455	12/2007	Harris et al.	N/A	N/A
2009/0005863	12/2008	Goetz et al.	N/A	N/A
2009/0024110	12/2008	Heideman et al.	N/A	N/A
2009/0043382	12/2008	Maurer et al.	N/A	N/A
2009/0131880	12/2008	Speziali et al.	N/A	N/A
2009/0156995	12/2008	Martin et al.	N/A	N/A
2009/0163934	12/2008	Raschdorf, Jr. et al.	N/A	N/A
2009/0166913	12/2008	Guo et al.	N/A	N/A
2009/0177266	12/2008	Powell et al.	N/A	N/A
2009/0234280	12/2008	Tah et al.	N/A	N/A
2009/0275902	12/2008	Heeps et al.	N/A	N/A
2009/0287304	12/2008	Dahlgren et al.	N/A	N/A
2010/0022823	12/2009	Goldfarb et al.	N/A	N/A
2010/0057192	12/2009	Celermajer	N/A	N/A
2010/0069834	12/2009	Schultz	N/A	N/A
2010/0094317	12/2009	Goldfarb et al.	N/A	N/A
2010/0106141	12/2009	Osypka et al.	N/A	N/A
2010/0121434	12/2009	Paul et al.	N/A	N/A
2010/0249497	12/2009	Peine et al.	N/A	N/A
2010/0298929	12/2009	Thornton et al.	N/A	N/A
2010/0324595	12/2009	Linder et al.	N/A	N/A
2011/0082538	12/2010	Dahlgren et al.	N/A	N/A
2011/0137410	12/2010	Hacohen	N/A	N/A
2011/0245855	12/2010	Matsuoka et al.	N/A	N/A
2011/0257723	12/2010	McNamara	N/A	N/A
2011/0295281	12/2010	Mizumoto et al.	N/A	N/A
2012/0022633	12/2011	Olson et al.	N/A	N/A
2012/0089125	12/2011	Scheibe et al.	N/A	N/A
2012/0109160	12/2011	Martinez et al.	N/A	N/A
2012/0116419	12/2011	Sigmon, Jr.	N/A	N/A
2012/0209318	12/2011	Qadeer	N/A	N/A
2012/0277853	12/2011	Rothstein	N/A	N/A
2013/0035759	12/2012	Gross et al.	N/A	N/A

2013/0041314	12/2012	Dillon	N/A	N/A
2013/0066341	12/2012	Ketai et al.	N/A	N/A
2013/0066342	12/2012	Dell et al.	N/A	N/A
2013/0072945	12/2012	Terada	N/A	N/A
2013/0073034	12/2012	Wilson et al.	N/A	N/A
2013/0110254	12/2012	Osborne	N/A	N/A
2013/0190798	12/2012	Kapadia	N/A	N/A
2013/0190861	12/2012	Chau et al.	N/A	N/A
2013/0268069	12/2012	Zakai et al.	N/A	N/A
2013/0282059	12/2012	Ketai et al.	N/A	N/A
2013/0304197	12/2012	Buchbinder et al.	N/A	N/A
2013/0325110	12/2012	Khalil et al.	N/A	N/A
2014/0031928	12/2013	Murphy et al.	N/A	N/A
2014/0046433	12/2013	Kovalsky	N/A	N/A
2014/0046434	12/2013	Rolando et al.	N/A	N/A
2014/0052237	12/2013	Lane et al.	N/A	N/A
2014/0058411	12/2013	Soutorine et al.	N/A	N/A
2014/0067048	12/2013	Chau et al.	N/A	N/A
2014/0067052	12/2013	Chau et al.	N/A	N/A
2014/0094903	12/2013	Miller et al.	N/A	N/A
2014/0135685	12/2013	Kabe et al.	N/A	N/A
2014/0194975	12/2013	Quill et al.	N/A	N/A
2014/0200662	12/2013	Eftel et al.	N/A	N/A
2014/0207231	12/2013	Hacohen et al.	N/A	N/A
2014/0222136	12/2013	Geist et al.	N/A	N/A
2014/0236198	12/2013	Goldfarb et al.	N/A	N/A
2014/0243968	12/2013	Padala	N/A	N/A
2014/0251042	12/2013	Asselin et al.	N/A	N/A
2014/0277404	12/2013	Wilson et al.	N/A	N/A
2014/0277411	12/2013	Bortlein et al.	N/A	N/A
2014/0277427	12/2013	Ratz et al.	N/A	N/A
2014/0316428	12/2013	Golan	N/A	N/A
2014/0324164	12/2013	Gross et al.	N/A	N/A
2014/0330368	12/2013	Gloss et al.	N/A	N/A
2014/0336751	12/2013	Kramer	N/A	N/A
2014/0371843	12/2013	Wilson et al.	N/A	N/A
2015/0039084	12/2014	Levi et al.	N/A	N/A
2015/0057704	12/2014	Takahashi	N/A	N/A
2015/0094802	12/2014	Buchbinder et al.	N/A	N/A
2015/0100116	12/2014	Mohl et al.	N/A	N/A
2015/0105808	12/2014	Gordon et al.	N/A	N/A
2015/0112429	12/2014	Khairkhahan et al.	N/A	N/A
2015/0119981	12/2014	Khairkhahan et al.	N/A	N/A
2015/0148896	12/2014	Karapetian et al.	N/A	N/A
2015/0157268	12/2014	Winshtein et al.	N/A	N/A
2015/0196390	12/2014	Ma et al.	N/A	N/A
2015/0223793	12/2014	Goldfarb et al.	N/A	N/A
2015/0230919	12/2014	Chau et al.	N/A	N/A
2015/0238313	12/2014	Spence et al.	N/A	N/A
2015/0257757	12/2014	Powers et al.	N/A	N/A

2015/0257877	12/2014	Hernandez	N/A	N/A
2015/0257883	12/2014	Basude et al.	N/A	N/A
2015/0313592	12/2014	Coillard-Lavirotte et al.	N/A	N/A
2015/0351904	12/2014	Cooper et al.	N/A	N/A
2015/0366666	12/2014	Khairkhahan et al.	N/A	N/A
2016/0008129	12/2015	Siegel	N/A	N/A
2016/0008131	12/2015	Christianson et al.	N/A	N/A
2016/0022970	12/2015	Forcucci et al.	N/A	N/A
2016/0038280	12/2015	Morriss	623/2.18	A61F 2/2436
2016/0051796	12/2015	Kanemasa et al.	N/A	N/A
2016/0074164	12/2015	Naor	N/A	N/A
2016/0074165	12/2015	Spence et al.	N/A	N/A
2016/0106539	12/2015	Buchbinder et al.	N/A	N/A
2016/0113762	12/2015	Clague et al.	N/A	N/A
2016/0113764	12/2015	Sheahan et al.	N/A	N/A
2016/0113766	12/2015	Ganesan et al.	N/A	N/A
2016/0155987	12/2015	Yoo et al.	N/A	N/A
2016/0166382	12/2015	Nguyen	N/A	N/A
2016/0174979	12/2015	Wei	N/A	N/A
2016/0174981	12/2015	Fago et al.	N/A	N/A
2016/0199181	12/2015	Kramer	N/A	N/A
2016/0242906	12/2015	Morriss et al.	N/A	N/A
2016/0287387	12/2015	Wei	N/A	N/A
2016/0302811	12/2015	Rodriguez-Navarro et al.	N/A	N/A
2016/0317290	12/2015	Chau et al.	N/A	N/A
2016/0331523	12/2015	Chau et al.	N/A	N/A
2016/0354082	12/2015	Oz et al.	N/A	N/A
2017/0020521	12/2016	Krone et al.	N/A	N/A
2017/0035561	12/2016	Rowe et al.	N/A	N/A
2017/0035566	12/2016	Krone et al.	N/A	N/A
2017/0042456	12/2016	Budiman	N/A	N/A
2017/0042678	12/2016	Ganesan et al.	N/A	N/A
2017/0049455	12/2016	Seguin	N/A	N/A
2017/0100119	12/2016	Baird et al.	N/A	N/A
2017/0100236	12/2016	Robertson et al.	N/A	N/A
2017/0224955	12/2016	Douglas et al.	N/A	N/A
2017/0239048	12/2016	Goldfarb et al.	N/A	N/A
2017/0252154	12/2016	Tubishevitz et al.	N/A	N/A
2017/0266413	12/2016	Khuu et al.	N/A	N/A
2017/0281330	12/2016	Liljegren et al.	N/A	N/A
2017/0348102	12/2016	Cousins et al.	N/A	N/A
2018/0008311	12/2017	Shiroff et al.	N/A	N/A
2018/0021044	12/2017	Miller et al.	N/A	N/A
2018/0021129	12/2017	Peterson et al.	N/A	N/A
2018/0021134	12/2017	McNiven et al.	N/A	N/A
2018/0071487	12/2017	Khuu et al.	N/A	N/A
2018/0078271	12/2017	Thrasher, III	N/A	N/A
2018/0078361	12/2017	Naor et al.	N/A	N/A

2018/0092661	12/2017	Prabhu	N/A	N/A
2018/0126119	12/2017	McNiven et al.	N/A	N/A
2018/0126124	12/2017	Winston et al.	N/A	N/A
2018/0133008	12/2017	Kizuka et al.	N/A	N/A
2018/0146964	12/2017	Garcia et al.	N/A	N/A
2018/0146966	12/2017	Hernandez et al.	N/A	N/A
2018/0153552	12/2017	King et al.	N/A	N/A
2018/0161159	12/2017	Lee et al.	N/A	N/A
2018/0168803	12/2017	Pesce et al.	N/A	N/A
2018/0185154	12/2017	Cao	N/A	N/A
2018/0221147	12/2017	Ganesan et al.	N/A	N/A
2018/0235657	12/2017	Abunassar	N/A	N/A
2018/0243086	12/2017	Barbarino et al.	N/A	N/A
2018/0258665	12/2017	Reddy et al.	N/A	N/A
2018/0263767	12/2017	Chau et al.	N/A	N/A
2018/0296326	12/2017	Dixon et al.	N/A	N/A
2018/0296327	12/2017	Dixon et al.	N/A	N/A
2018/0296328	12/2017	Dixon et al.	N/A	N/A
2018/0296329	12/2017	Dixon et al.	N/A	N/A
2018/0296330	12/2017	Dixon et al.	N/A	N/A
2018/0296331	12/2017	Dixon et al.	N/A	N/A
2018/0296332	12/2017	Dixon et al.	N/A	N/A
2018/0296333	12/2017	Dixon et al.	N/A	N/A
2018/0296334	12/2017	Dixon et al.	N/A	N/A
2018/0325661	12/2017	Delgado et al.	N/A	N/A
2018/0325671	12/2017	Abunassar et al.	N/A	N/A
2018/0333259	12/2017	Dibie	N/A	N/A
2018/0344457	12/2017	Gross et al.	N/A	N/A
2018/0353181	12/2017	Wei	N/A	N/A
2019/0000613	12/2018	Delgado et al.	N/A	N/A
2019/0000623	12/2018	Pan et al.	N/A	N/A
2019/0008642	12/2018	Delgado et al.	N/A	N/A
2019/0008643	12/2018	Delgado et al.	N/A	N/A
2019/0015199	12/2018	Delgado et al.	N/A	N/A
2019/0015200	12/2018	Delgado et al.	N/A	N/A
2019/0015207	12/2018	Delgado et al.	N/A	N/A
2019/0015208	12/2018	Delgado et al.	N/A	N/A
2019/0021851	12/2018	Delgado et al.	N/A	N/A
2019/0021852	12/2018	Delgado et al.	N/A	N/A
2019/0029498	12/2018	Mankowski et al.	N/A	N/A
2019/0029810	12/2018	Delgado et al.	N/A	N/A
2019/0029813	12/2018	Delgado et al.	N/A	N/A
2019/0030285	12/2018	Prabhu et al.	N/A	N/A
2019/0053810	12/2018	Griffin	N/A	N/A
2019/0060058	12/2018	Delgado et al.	N/A	N/A
2019/0060059	12/2018	Delgado et al.	N/A	N/A
2019/0060072	12/2018	Zeng	N/A	N/A
2019/0060073	12/2018	Delgado et al.	N/A	N/A
2019/0060074	12/2018	Delgado et al.	N/A	N/A
2019/0060075	12/2018	Delgado et al.	N/A	N/A

2019/0069991	12/2018	Metchik et al.	N/A	N/A
2019/0069992	12/2018	Delgado et al.	N/A	N/A
2019/0069993	12/2018	Delgado et al.	N/A	N/A
2019/0105156	12/2018	He et al.	N/A	N/A
2019/0111239	12/2018	Bolduc et al.	N/A	N/A
2019/0117113	12/2018	Curran	N/A	N/A
2019/0142589	12/2018	Basude	N/A	N/A
2019/0159782	12/2018	Kamaraj et al.	N/A	N/A
2019/0167197	12/2018	Abunassar et al.	N/A	N/A
2019/0183644	12/2018	Hacohen	N/A	N/A
2019/0192296	12/2018	Schwartz et al.	N/A	N/A
2019/0209323	12/2018	Metchik et al.	N/A	N/A
2019/0261995	12/2018	Goldfarb et al.	N/A	N/A
2019/0261996	12/2018	Goldfarb et al.	N/A	N/A
2019/0261997	12/2018	Goldfarb et al.	N/A	N/A
2019/0314155	12/2018	Franklin et al.	N/A	N/A
2019/0321166	12/2018	Freschauf et al.	N/A	N/A
2020/0113683	12/2019	Dale et al.	N/A	N/A
2020/0138569	12/2019	Basude et al.	N/A	N/A
2020/0205979	12/2019	O'Carroll et al.	N/A	N/A
2020/0237512	12/2019	McCann et al.	N/A	N/A
2020/0337842	12/2019	Metchik et al.	N/A	N/A
2020/0352717	12/2019	Kheradvar et al.	N/A	N/A
2020/0360054	12/2019	Walsh et al.	N/A	N/A
2020/0360132	12/2019	Spence	N/A	N/A
2020/0368016	12/2019	Pesce et al.	N/A	N/A
2021/0022850	12/2020	Basude et al.	N/A	N/A
2021/0059680	12/2020	Lin et al.	N/A	N/A
2021/0169650	12/2020	Dai et al.	N/A	N/A
2021/0186698	12/2020	Abunassar et al.	N/A	N/A
2021/0251757	12/2020	Siegel et al.	N/A	N/A
2021/0259835	12/2020	Tyler, II et al.	N/A	N/A
2021/0267781	12/2020	Metchik et al.	N/A	N/A
2021/0307900	12/2020	Hacohen	N/A	N/A
2021/0330456	12/2020	Hacohen et al.	N/A	N/A
2021/0338418	12/2020	Feld	N/A	N/A
2021/0361238	12/2020	Bak-Boychuk et al.	N/A	N/A
2021/0361416	12/2020	Stearns	N/A	N/A
2021/0361422	12/2020	Gross et al.	N/A	N/A
2021/0361428	12/2020	Dixon	N/A	N/A
2021/0378818	12/2020	Manash et al.	N/A	N/A
2021/0401434	12/2020	Tien et al.	N/A	N/A
2022/0039943	12/2021	Phan	N/A	N/A
2022/0039954	12/2021	Nia et al.	N/A	N/A
2022/0071767	12/2021	Dixon et al.	N/A	N/A
2022/0133327	12/2021	Zhang et al.	N/A	N/A
2022/0142780	12/2021	Zhang et al.	N/A	N/A
2022/0142781	12/2021	Zhang et al.	N/A	N/A
2022/0226108	12/2021	Freschauf et al.	N/A	N/A
2022/0233312	12/2021	Delgado et al.	N/A	N/A

2022/0257196	12/2021	Massmann	N/A	N/A
2022/0287841	12/2021	Freschauf et al.	N/A	N/A
2022/0296248	12/2021	Abunassar et al.	N/A	N/A
2022/0313433	12/2021	Ma et al.	N/A	N/A
2023/0014540	12/2022	Metchik et al.	N/A	N/A
2023/0149170	12/2022	Giese et al.	N/A	N/A
2023/0218291	12/2022	Zarbatany et al.	N/A	N/A
2023/0270549	12/2022	Guidotti et al.	N/A	N/A
2024/0148505	12/2023	Datta et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
1142351	12/1996	CN	N/A
106175845	12/2015	CN	N/A
106491245	12/2016	CN	N/A
107789017	12/2017	CN	N/A
109953779	12/2018	CN	N/A
110338857	12/2018	CN	N/A
110495972	12/2018	CN	N/A
110537946	12/2018	CN	N/A
110664515	12/2019	CN	N/A
209996540	12/2019	CN	N/A
211243911	12/2019	CN	N/A
211723546	12/2019	CN	N/A
111870398	12/2019	CN	N/A
111904660	12/2019	CN	N/A
112120831	12/2019	CN	N/A
112168427	12/2020	CN	N/A
112190367	12/2020	CN	N/A
212346813	12/2020	CN	N/A
212415988	12/2020	CN	N/A
212490263	12/2020	CN	N/A
113476182	12/2020	CN	N/A
113855328	12/2020	CN	N/A
215019733	12/2020	CN	N/A
0098100	12/1983	EP	N/A
2146050	12/1972	FR	N/A
9711600	12/1996	FR	N/A
2006032051	12/2005	WO	N/A
2014064694	12/2013	WO	N/A
2016183485	12/2015	WO	N/A
2017015632	12/2016	WO	N/A
2018013856	12/2017	WO	N/A
2018050200	12/2017	WO	N/A
2018050203	12/2017	WO	N/A
2018195015	12/2017	WO	N/A
2018195201	12/2017	WO	N/A
2018195215	12/2017	WO	N/A
2019139904	12/2018	WO	N/A
2020106705	12/2019	WO	N/A

2020106827	12/2019	WO	N/A
2020112622	12/2019	WO	N/A
2020167677	12/2019	WO	N/A
2020168081	12/2019	WO	N/A
2020172224	12/2019	WO	N/A
2020176410	12/2019	WO	N/A
2021196580	12/2020	WO	N/A
2021227412	12/2020	WO	N/A
2022006087	12/2021	WO	N/A
2022036209	12/2021	WO	N/A
2022051241	12/2021	WO	N/A
2022052506	12/2021	WO	N/A
2022068188	12/2021	WO	N/A
2022101817	12/2021	WO	N/A
2022140175	12/2021	WO	N/A
2022153131	12/2021	WO	N/A
2022155298	12/2021	WO	N/A
2022157592	12/2021	WO	N/A
2022212172	12/2021	WO	N/A
2023003755	12/2022	WO	N/A
2023004098	12/2022	WO	N/A
2023278663	12/2022	WO	N/A
2023288003	12/2022	WO	N/A

OTHER PUBLICATIONS

Al Zaibag et al., “Percutaneous Balloon Valvotomy in Tricuspid Stenosis”, British Heart Journal, vol. 57, No. 1, Jan. 1987. cited by applicant

Al-Khaja et al., “Eleven Years' Experience with Carpentier-Edwards Biological Valves in Relation to Survival and Complications”, European Journal of Cardio-thoracic Surgery 3: pp. 305-311, 1989. cited by applicant

Almagor et al., “Balloon Expandable Stent Implantation in Stenotic Right Heart Valved Conduits”, Journal of the American College of Cardiology, vol. 16, No. 6, pp. 1310-1314, Nov. 15, 1990. cited by applicant

Andersen, et al., “Transluminal implantation of artificial heart valves. Description of a new expandable aortic valve and initial results with implantation by catheter technique in closed chest pigs.” European Heart Journal (1992), 13, 704-708. cited by applicant

Andersen, H.R. “History of Percutaneous Aortic Valve Prosthesis,” Herz No. 34. pp. 343-346. 2009. cited by applicant

Batista RJ et al., “Partial left ventriculectomy to treat end-stage heart disease”, Ann Thorac Surg., vol. 64, Issue-3, pp. 634-638, Sep. 1997. cited by applicant

Beall AC Jr. et al., “Clinical experience with a dacron velour-covered teflon-disc mitral-valve prosthesis”, Ann Thorac Surg., vol. 5, Issue 5, pp. 402-410, May 1968. cited by applicant

Benchimol et al., “Simultaneous Left Ventricular Echocardiography and Aortic Blood Velocity During Rapid Right Ventricular Pacing in Man”, The American Journal of the Medical Sciences, vol. 273, No. 1, pp. 55-62, 1977. cited by applicant

Dake et al., “Transluminal Placement of Endovascular Stent-Grafts for the Treatment of Descending Thoracic Aortic Aneurysms”, The New England Journal of Medicine, vol. 331, No. 26, pp. 1729-1734, Dec. 29, 1994. cited by applicant

Dotter et al., “Transluminal Treatment of Arteriosclerotic Obstruction: Description of a New Technic and a Preliminary Report of Its Application”, Circulation, vol. XXX, No. 30, pp. 654-670,

Nov. 1, 1964, Lippincott Williams & Wilkins, Philadelphia, PA. cited by applicant
Fucci et al., “Improved results with mitral valve repair using new surgical techniques”, Eur J Cardiothorac Surg. 1995;Issue 9, vol. 11, pp. 621-626. cited by applicant
Inoune, M.D., Kanji, et al., “Clinical Application of Transvenous Mitral Commissurotomy by a New Balloon Catheter,” The Journal of Thoracic and Cardiovascular Surgery 87:394-402, 1984. cited by applicant
Kolata, Gina “Device That Opens Clogged Arteries Gets a Failing Grade in a New Study”, The New York Times, Jan. 3, 1991, pp. 1-2 [online], [retrieved on Jul. 29, 2009]. Retrieved from the Internet <URL:http://www.nytimes.com/1991/01/03/health/device-that-opens-clogged-arteries-gets-a-faili . . . . cited by applicant
Lawrence, Jr., et al., “Percutaneous Endovascular Graft: Experimental Evaluation”, Cardiovascular Radiology 163, pp. 357-360, May 1987. cited by applicant
Maisano F et al., ‘The edge-to-edge technique: a simplified method to correct mitral insufficiency’, Eur J Cardiothorac Surg., vol. 13, Issue-3, pp. 240-245, Mar. 1998. cited by applicant
Pavcnik, M.D., Ph.D., Dusan, et al. “Development and Initial Experimental Evaluation of a Prosthetic Aortic Valve for Transcatheter Placement,” Cardiovascular Radiology 1992; 183: 151-154. cited by applicant
Porstmann et al., “Der Verschluß des Ductus Arteriosus Persistens Ohne Thorakotomie”, Thoraxchirurgie Vaskuläre Chirurgie, Band 15, Heft 2, Stuttgart, im Apr. 1967, pp. 199-203. cited by applicant
Praz et al., “Compassionate use of the PASCAL transcatheter mitral valve repair system for patients with severe mitral regurgitation: a multicentre, prospective, observational, first-in-man study,” Lancet vol. 390, pp. 773-780, 2017. cited by applicant
Rashkind et al., “Creation of an Atrial Septal Defect Without Thoracotomy: A Pallative Approach to Complete Transposition of the Great Arteries”, The Journal of the American Medical Association, vol. 196, No. 11, pp. 173-174, Jun. 13, 1956. cited by applicant
Rashkind et al., “Historical Aspects of Interventional Cardiology: Past, Present, and Future”, Texas Heart Institute Journal, Interventional Cardiology, pp. 363-367. cited by applicant
Reul RM et al., “Mitral valve reconstruction for mitral insufficiency”, Prog Cardiovasc Dis., vol. 39, Issue-6, May-Jun. 1997. cited by applicant
Rosch, M.D., Josef, “The Birth, Early Years and Future of Interventional Radiology,” J Vasc Interv Radiol 2003; 14:841-853. cited by applicant
Ross, D.N, “Aortic Valve Surgery”, Surgery of the Aortic Valves, Guy's Hospital, London, pp. 192-197. cited by applicant
Sabbah et al., “Mechanical Factors in the Degeneration of Porcine Bioprosthetic Valves: An Overview”, Journal of Cardiac Surgery, vol. 4, No. 4, pp. 302-309, Dec. 1989. cited by applicant
Selby et al., “Experience with New Retrieval Forceps for Foreign Body Removal in the Vascular, Urinary, and Biliary Systems”, Radiology: 176. pp. 535-538, 1990. cited by applicant
Serruys et al., “Stenting of Coronary Arteries. Are we the Sorcerer's Apprentice?”, European Heart Journal, 10, 774-782, pp. 37-45, 1989. cited by applicant
Sigwart, Ulrich, “An Overview of Intravascular Stents: Old and New,” Chapter 48, Textbook of Interventional Cardiology, 2nd Edition, W.B. Saunders Company, Philadelphia, PA, @ 1994, 1990, pp. 803-815. cited by applicant
Uchida et al., “Modifications of Gianturco Expandable Wire Stents”, Technical Note, American Roentgen Ray Society, pp. 1185-1187, May 1988. cited by applicant
Umaña JP et al., Bow-tie' mitral valve repair: an adjuvant technique for ischemic mitral regurgitation, Ann Thorac Surg., vol. 66, Issue-6, pp. 1640-1646, Nov. 1998. cited by applicant
“Urban, Philip MD, ““Coronary Artery Stenting””, pp. 5-47, @ 1991, ISBN: 2-88049-054-5, Editions Medecine et Hygiene, Geneva, Switzerland.” cited by applicant
Watt et al., “Intravenous Adenosine in the Treatment of Supraventricular Tachycardia: A Dose-

Ranging Study and Interaction with Dipyridamole”, Br. J. Clin. Pharmac. 21, pp. 227-230, 1986. cited by applicant
Wheatley, David J., “Valve Prosthesis”, Rob & Smith's Operative Surgery, pp. 415-424, 1986. cited by applicant
Grasso et al., “The PASCAL transcatheter mitral valve repair system for the treatment ofanother piece to the puzzle of edge-to-edge technique”, Journal of Thoracic Disease, vol. 9, No. 12, pp. 4856-4859, Dec. 2017, doi:10.21037/jtd.2017.10.156, AME Publishing Company, Hong Kong, China. cited by applicant

---

Primary Examiner: Edwards; Jerrah

Assistant Examiner: Trevino, III; Jose H.

Background/Summary

RELATED PATENT APPLICATIONS (1) This patent application is a continuation of U.S. patent application Ser. No. 16/121,507, filed on Sep. 4, 2018, which claims priority to U.S. Provisional Application No. 62/555,863, filed Sep. 8, 2017, and which is related to co-pending U.S. patent application Ser. No. 16/112,388, filed Aug. 24, 2018 and entitled “Transcatheter Device for Treating Mitral Regurgitation,” the disclosures of all of these are incorporated by reference herein in their entireties for all purposes.

FIELD OF THE INVENTION
(1) The present invention relates to the repair of heart valves, and, more particularly, to methods and apparatuses for the repair of heart valves by positioning a device between valve leaflets to improve valve closure.
BACKGROUND OF THE INVENTION
(2) In vertebrate animals, the heart is a hollow muscular organ having four pumping chambers: the left and right atria and the left and right ventricles, each provided with its own one-way outflow valve. The natural heart valves are identified as the aortic, mitral (or bicuspid), tricuspid and pulmonary valves. The valves separate the chambers of the heart, and are each mounted in an annulus therebetween. The annuluses comprise dense fibrous rings attached either directly or indirectly to the atrial and ventricular muscle fibers. The leaflets are flexible collagenous structures that are attached to and extend inward from the annuluses to meet at coapting edges. The aortic, tricuspid, and pulmonary valves usually have three leaflets, while the mitral valve usually has two leaflets.
(3) The operation of the heart, and thus the patient's health, may be seriously impaired if any of the heart valves is not functioning properly. Various problems can develop with heart valves for a number of clinical reasons. Stenosis in heart valves is a condition in which the valves do not open properly. Insufficiency is a condition which a valve does not close properly. Repair or replacement of the aortic or mitral valves are most common because they reside in the left side of the heart where pressures and stresses are the greatest. In a valve replacement operation, a replacement prosthetic valve is implanted into the native valve annulus, which may involve excision of the native valve leaflets.
(4) In many patients who suffer from valve dysfunction, surgical or percutaneous repair (i.e., “valvuloplasty”) is a desirable alternative to valve replacement. Remodeling of the valve annulus (i.e., “annuloplasty”) is central to many reconstructive valvuloplasty procedures. Remodeling of the valve annulus is typically accomplished by implantation of a prosthetic ring (i.e. “annuloplasty ring”) to stabilize the annulus and to correct or prevent valvular insufficiency that may result from

a dysfunction of the valve annulus. Annuloplasty rings are typically constructed of a resilient core covered with a fabric sewing ring. Annuloplasty procedures are performed not only to repair damaged or diseased annuli, but may also be performed in conjunction with other procedures, such as leaflet repair.

(5) Heart valves may lose their ability to close properly due to dilation of an annulus around the valve or a flaccid, prolapsed leaflet. The leaflets may also have shrunk due to disease, such as rheumatic disease, thereby leaving a gap in the valve between the leaflets. The inability of the heart valve to close will cause blood to leak backwards (opposite to the normal flow of blood), commonly referred to as regurgitation. Common examples of such regurgitation include mitral valve regurgitation (i.e., leakage of blood through the mitral valve and back into the left atrium) and aortic valve regurgitation (i.e., leakage through the aortic valve back into the left ventricle). Regurgitation may seriously impair the function of the heart since more blood will have to be pumped through the regurgitating valve to maintain adequate circulation.

(6) Heart valve regurgitation decreases the efficiency of the heart, reduces blood circulation, and adds stress to the heart. In early stages, heart valve regurgitation leaves a person fatigued and short of breath. If left unchecked, the problem can lead to congestive heart failure, arrhythmias, or death.

(7) Mitral valve regurgitation may be caused by dysfunction of the mitral valve structure, such as may result from direct injury to the mitral valve leaflets. Such regurgitation can be caused by changes in the shape of the mitral valve annulus, damage to the posterior and/or anterior leaflets, and/or damage to the chordae tendinae. In such regurgitation, the anterior and posterior leaflets no longer coapt together properly to seal the valve, so that instead of the anterior and posterior leaflets coapting to fully close the mitral valve annulus during systole, an opening remains between the edges of the anterior and posterior leaflets.

(8) Various methods of mitral valve repair are known in the art. Implantation of an annuloplasty ring, typically around the posterior aspect of the mitral valve, has proven successful in a number of cases. Such annuloplasty rings reshape the surrounding annulus, which can lead to proper coaptation of the native leaflets. Another repair technique for the mitral valve is known as a “bow-tie” repair, which involves suturing the anterior and posterior leaflets together in edge-to-edge fashion toward the middle of the leaflets, causing blood to flow through the two side openings thus formed. This process was originally developed by Dr. Ottavio Alfieri, and involved placing the patient on extracorporeal bypass in order to access and suture the mitral valve leaflets. Later adaptations of the bow-tie technique involved beating-heart repairs using percutaneous methods, such using a catheter to install suture or a clip to secure the opposing leaflets together.

(9) Another approach to repairing a native valve having non-coapting leaflets, including mitral and aortic valves, involves inserting a device between the leaflets, with the device being sized and positioned to block the gap between the otherwise non-coapting leaflets. Examples of such repair devices and techniques are disclosed in U.S. Pat. No. 8,968,395 to Hauser et al. and U.S. Patent Pub. No. 2009/0043382 for Maurer et al. These disclose devices which include an anchor deployed in the lower ventricle which secures a blocking device within the mitral valve annulus.

(10) There is presently a need for an improved means for performing heart valve repair. The current invention fulfills this need.

SUMMARY OF THE INVENTION

(11) The present invention provides a number of devices and methods for improving valve function. The devices and methods herein reduce or eliminate valve regurgitation without interfering with normal valve function, i.e., not impeding the natural motion of the leaflets, chordae tendinae, or papillary muscles.

(12) It should be understood that each of the sealing elements disclosed herein can be used with any and all of the anchor elements disclosed herein, even though the specific combination of sealing element with anchor elements may not be explicitly shown in the figures herein. In other words, based on the explanation of the particular device, one of skill in the art should have little

trouble combining the features of certain of two such devices. Therefore, it should be understood that many of the sealing and anchor elements are interchangeable, and the invention covers all permutations thereof. Moreover, each of the sealing elements disclosed herein may be used alone or in combination with other anchor devices, and each of the anchor elements disclosed herein can be used alone or in combination with other implant devices, such as anchor and sealing elements disclosed in U.S. patent application Ser. No. 16/112,388, filed Aug. 24, 2018 and entitled "Transcatheter Device for Treating Mitral Regurgitation."

(13) The devices of the present invention can be utilized in standard open surgical procedures, minimally-invasive procedures, or percutaneous procedures. In one embodiment the devices can be delivered through an open chest, e.g., transapically or transatrially. In another embodiment, the devices can be introduced through an incision performed over the roof of the left atrium. In yet another embodiment the devices can be delivered into the left ventricle through the right chest via a thorascope, which may be performed transapically. The devices can also be delivered percutaneously, such as via a catheter or catheters into the patient's arterial system (e.g. through the femoral or brachial arteries).

(14) Advantages of the device include a low delivery profile, which is conducive to minimally-invasive and percutaneous delivery methods. The device is configured to interact properly with the native leaflets, ventricle, atrial, and subvalvular apparatus. The device preserves rather than obstructs the mobility and dynamic motion of the native leaflets (except as necessary for proper coaptation). The native leaflets and chordae tendinae are preserved, and continue to operate (including opposing the systolic closing pressure). The subvalvular process and left ventricle coordination are thus preserved. The device may be configured so that it does not expand the native mitral valve leaflets or annulus outward, so that left ventricular outflow tract (LVOT) impingement/obstruction should not be a concern. The shape and low profile of the sealing element minimizes flow resistance during diastole, and there are no areas of stasis created by the device. A single device can be applicable to a wide range of valve sizes.

(15) The device may be applicable to numerous mitral valve regurgitation conditions, including those caused by leaflet prolapse with varying amounts of annular dilatation (type I), focal leaflet prolapse (type II), and leaflet tethering (type IIIb).

(16) Embodiments of the present disclosure provide devices and methods for improving the function of a defective heart valve, such as a mitral valve. The devices and methods disclosed herein are desirably delivered into a subject's heart using percutaneous or minimally invasive surgical methods. Accordingly, desirable delivery methods described herein may not require extracorporeal circulation (e.g., blood from a subject's circulation being routed outside the body to have a process applied to and then, returned of the subject's circulation). For example, in one embodiment, a delivery catheter (or similar delivery device) is inserted through an incision in the chest wall and then through the cardiac tissue (e.g., through the apex of the heart) into a chamber of the patient's beating heart. The delivery catheter can allow a prosthetic device to be delivered into the heart in a collapsed configuration and then expanded within the heart for treating a defective heart valve. Because delivery methods may not require extracorporeal circulation, complications can be greatly reduced as compared with traditional open-heart surgery.

(17) An embodiment of the invention for treating a mitral valve is a device that includes an expandable prosthetic sealing member having an axisymmetrical top profile (or an elongated or elliptical top profile) when expanded, the sealing member shaped when expanded for contacting the leaflets of the mitral valve. The device also includes an anchoring member coupled to the sealing member and configured to secure the sealing member at a desired position between the mitral valve leaflets. The anchor member may have an axisymmetrical top profile. The sealing member and anchoring member may be radially collapsible and radially expandable, which may permit the device to be delivered and deployed via a catheter.

(18) Various anchoring elements are within the scope of the invention. Many of the anchoring

elements may be axisymmetric, which is to say symmetrical about an axis running from a lower (e.g., ventricular) end to an upper (e.g., atrial) end. In one embodiment, an anchor element may have an upper portion configured to extend around a mitral valve annulus and contact atrial tissue adjacent the mitral valve annulus, a lower portion configured to extend around native valve leaflets and engage ventricular tissue adjacent the mitral valve annulus without interfering with the movement of the mitral valve leaflets, and a central portion configured to support the sealing member. The anchor element upper portion may have an upper portion may have a plurality of radially-extending arms to engage heart tissue such as atrial tissue adjacent a mitral valve. The lower portion may have a plurality of radially-extending arms to engage heart tissue such as ventricular tissue adjacent a mitral valve, and may be dimensioned such when the lower portion is deployed the native valve leaflets can open and close as the heart beats with limited or no interference from the lower portion. In one embodiment of the device, the native valve leaflets are unrestricted in their opening and closing by any and all portions of the device except for the sealing element, which engages the native valve leaflets during systole to form a seal between the native leaflets and thereby prevent mitral valve regurgitation.

(19) An anchoring member according to the invention may be configured so that the anchoring member does not expand the native mitral valve annulus, because such annulus expansion might otherwise cause reduction in valve efficiency. For example, the anchoring member may be configured so it does not subject the native mitral valve annulus to radially-expansive forces. The anchoring member may be configured so that the lower anchor portion presses upward against the native valve annulus while the upper anchor portion presses downward against the native valve annulus, so that tissue of the native valve annulus is held between the lower anchor portion and the upper anchor portion but the annulus is not subjected to radially-expansive forces from the anchoring member. Such embodiments may even prevent further annulus expansion by securing the annulus between the opposing anchor portions.

(20) An anchoring member according to the invention may be self-expandable, such as via construction of a memory material such as nitinol. The anchoring member may alternatively be formed of other materials, such as stainless steel or cobalt chromium.

(21) Another anchor element according to the invention is configured for deployment in a single heart chamber such as the atrium, with only the sealing element extending out of the chamber and into the heart valve and annulus. For example, an anchor element may have a plurality of upper arms configured to engage the upper portion of the heart chamber, and a plurality of lower arms configured to engage the lower portion of that same heart chamber. The upper or lower arms, or other structure of the anchor element, may have curves configured to act as shock absorbers to permit the anchor element to flex responsive to movements of the heart chamber as the heart beats.

(22) A sealing element according to the invention is configured to be introduced in a radially collapsed but lengthened configuration, and then be shortened and radially expanded to a desired shape for improving valve function. The sealing element may be axisymmetric or elongated (e.g., elliptical) in top profile, and may dimensioned to be deployed in an annulus of a native valve (e.g., mitral valve) of a heart at a position between native valve leaflets to contact the native valve leaflets during ventricular systole to create a seal to prevent regurgitation of blood from the ventricle to the atrium, while permitting the native valve leaflets to open and close as the heart beats. The sealing element may have an upper end, a lower end, an anterior surface, and a posterior surface. The anterior surface may be configured to coapt with a mitral valve anterior leaflet, and the posterior surface may configured to coapt with a mitral valve posterior leaflet. The sealing element may have a mesh support frame with a delivery configuration where the mesh support frame is substantially tubular with a delivery diameter, and an expanded configuration where the mesh support frame is radially expanded with an expanded central diameter at a center portion thereof while end portions of the mesh support frame remain in the delivery diameter. The expanded central diameter may be at least twice the delivery diameter, at least three times the delivery

diameter, at least four times the delivery diameter, at least five times the delivery diameter, etc. A covering may cover the mesh support frame to prevent the passage of blood therethrough. The sealing element in the expanded configuration may comprise an axisymmetrical, elongated, or elliptical top profile.
(23) A sealing element outer covering may preferably wrap around the exterior and/or interior of the central anchor portion or any other support structure for the sealing element, so that the wireform elements of the central anchor portion or other support frame are covered and/or encapsulated by the sealing element in order to prevent the native valve leaflets from contacting any frame elements of the central anchor portion.
(24) A system for treating a mitral valve according to an embodiment of the invention may have a delivery catheter, an anchor member, and a prosthetic sealing member. The anchoring member may be self-expanding and/or axisymmetric, and may have a plurality of radially-extendable arms for engaging heart tissue. The sealing member may be adapted for plugging a gap between the leaflets of the mitral valve and reducing regurgitation. The prosthetic sealing member may have a collapsed state and an expanded state, where in the collapsed state the sealing member is longer and thinner than in the expanded state. The sealing member may have an outer surface formed with biological tissue. The elongated cross-sectional profile of the sealing member may be solid such that blood is forced to flow around the sealing member.
(25) A method according to the invention for improving the function of a heart valve may involve advancing a distal end of a delivery catheter to a position at a mitral or other heart valve of a patient, wherein within the distal end is a prosthetic device having an anchor member and a sealing member. The sealing member may be configured to expand into a configuration to reduce regurgitation through the mitral valve. The anchor member may have an upper portion configured to expand into engagement with atrial tissue, a central portion configured to expand to support the sealing member, and a lower portion configured to expand into engagement with ventricular tissue. The anchor member may have an upper portion to engage the upper portion of the heart chamber, and a lower portion configured to engage the lower portion of that same heart chamber. The anchor member may be self-expandable, and/or formed of memory material, and may be mounted in a compressed state within a distal end of the delivery catheter. The method may further include releasing the anchor member upper portion from the catheter at a position such that the anchor member upper portion engages desired heart tissue, releasing the sealing member from the catheter, and releasing the anchor member lower portion from the catheter at a position such that the anchor member lower portion engages desired heart tissue. This deployment procedure could be performed in different orders, such as releasing the anchor member lower portion first, then the sealing member, and then the upper portion; or releasing the sealing member before or after the release of the lower and upper anchor portions. After deployment of the anchor member portions and the sealing member, the sealing member should be positioned between leaflets of the mitral valve such that during systole the leaflets coapt against the sealing member. The length and width of the sealing member may be adjusted after initial deployment by a user to improve heart valve function. The device may be delivered and deployed using various delivery techniques, such as percutaneously or transapically to the mitral or other heart valve.
(26) Methods of the invention may include advancing a distal end of a delivery catheter to a position at a heart valve (e.g., mitral valve) in a heart of a patient, wherein within the distal end is a prosthetic device having an anchor member and a sealing member, the sealing member being configured to expand into an axisymmetrical or elongated-and-symmetrical (e.g., elliptical) configuration to engage native mitral valve leaflets during systole while still allowing the native mitral valve leaflets to open and close as the heart beats; releasing the anchor member portion from the catheter at a position in the heart at or adjacent the native valve annulus to engage heart tissue and anchor the anchor member within the heart; releasing the sealing member from the catheter; radially expanding the sealing member, wherein the sealing member as it expands shortens in

length while increasing in diameter, wherein the sealing member after expansion comprises an axisymmetrical or elongated/elliptical top profile. After release of the anchor member and the sealing member, the sealing member is positioned between leaflets of the mitral valve such that during systole the leaflets coapt against the sealing member and the leaflets can open and close as the heart beats.
(27) The sealing member after expansion may have a circular top profile, or an elongated top profile (such as an elliptical top profile). The sealing member may be rotated about its central axis and with respect to the native valve annulus to a desired rotational position wherein the sealing member is aligned within the valve annulus with native valve features to improve valve leaflet coaptation against the sealing member. This rotation may be selectively performed by a user, and the user may also lock the sealing member at the desired rotational position. For an elongated sealing element, rotating the sealing member about its central axis may involve aligning a major axis of the sealing member to be substantially parallel (within 10 degrees) of a line between commissures of a mitral valve.
(28) Deployment of the device may be performed responsive to feedback of valve performance monitoring and/or visualization techniques. For example, while expanding or rotating the sealing element, a user may monitor valve performance and set the final expansion configuration and/or rotational position of the sealing element in a configuration where valve performance is maximized based on the valve performance and/or visualization feedback.
(29) The anchor member may comprise an upper portion and a lower portion. A central portion may be included, and configured to support the sealing member. In one embodiment, the upper portion is configured to expand into engagement with atrial tissue, and the lower portion is configured to expand into engagement with ventricular tissue. Methods for deploying it may include releasing the upper portion into engagement with atrial tissue, releasing the lower portion into engagement with ventricular tissue. In one embodiment, the upper portion is configured to expand into engagement with upper atrial tissue, and the lower portion is configured to expand into engagement with lower atrial tissue adjacent the native valve annulus. Methods for deploying it may include releasing the upper portion into engagement with upper atrial tissue, and releasing the lower portion into engagement with lower atrial tissue. Releasing the anchor member upper portion from the catheter may occur prior to, simultaneously with, or after releasing the anchor member lower portion from the catheter.
(30) The device may be delivered using various approaches, including percutaneously or transapically through the subject's vasculature.
(31) Other objects, features, and advantages of the present invention will become apparent from a consideration of the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a cross-sectional view of a heart;
(2) FIGS. **2**A-**2**D depict side cross-sectional views of a heart with a repair device deployed therein according to an embodiment of the invention;
(3) FIGS. **3**A-**3**D depict side views of a sealing element in various stages of expansion according to an embodiment of the invention;
(4) FIGS. **3**E-**3**H depict top views of the sealing element in the various stages of expansion depicted in FIGS. **3**A-**3**D;
(5) FIGS. **4**A-**4**B depict side and perspective views, respectively, of a slotted-tube sealing element frame in compressed and expanded configurations, respectively, according to an embodiment of the invention;

(6) FIGS. **5**A-**5**C depict side views of a braided mesh sealing element in various stages of expansion according to an embodiment of the invention;
(7) FIGS. **6**A-**6**D depict top views of a sealing element in various stages of expansion according to an embodiment of the invention;
(8) FIGS. **7**A-**7**C depict perspective views of various mechanisms for controlling the expansion of sealing elements according to embodiments of the invention;
(9) FIG. **8**A depicts a perspective view of a device according to an embodiment of the invention;
(10) FIGS. **8**B-**8**C depict top and bottom views, respectively, of an anchor according to an embodiment of the invention;
(11) FIGS. **8**D-**8**G depict side views of heart chambers with the device of FIG. **8**A at various stages of deployment therein according to an embodiment of the invention;
(12) FIG. **9**A depicts a side view of a device according to an embodiment of the invention;
(13) FIGS. **9**B-**9**C depict top and bottom views, respectively, of a frame according to an embodiment of the invention;
(14) FIGS. **9**D-**9**G depict side views of heart chambers with the device of FIG. **9**A at various stages of deployment therein according to an embodiment of the invention;
(15) FIG. **10**A depicts a side view of a device according to an embodiment of the invention;
(16) FIGS. **10**B-**10**E depict side views of heart chambers with the device of FIG. **10** at various stages of deployment therein according to an embodiment of the invention;
(17) FIGS. **11**A-**11**C depict perspective views of an anchor frame at with various portions expanded according to an embodiment of the invention; and
(18) FIG. **11**D depicts a device having the anchor element of FIGS. **11**A-**11**C in an expanded configuration according to an embodiment of the invention.

DETAILED DESCRIPTION OF SEVERAL EMBODIMENTS

(19) A cross-sectional view of a human heart **10** is depicted in FIG. **1**. The heart **10** has a muscular heart wall **11**, an apex **19**, and four chambers: right atrium **12**; right ventricle **14**; left atrium **16**; and left ventricle **18**. Blood flow is controlled by four main valves: tricuspid valve **20**; pulmonary valve **22**; mitral valve **24**; and aortic valve **26**. Blood flows through the superior vena cava **28** and the inferior vena cava **30** into the right atrium **12** of the heart **10**. The right atrium **12** pumps blood through the tricuspid valve **20** (in an open configuration) and into the right ventricle **14**. The right ventricle **14** then pumps blood out through the pulmonary valve **22** and into the pulmonary artery **32** (which branches into arteries leading to the lungs), with the tricuspid valve **20** closed to prevent blood from flowing from the right ventricle **14** back into the right atrium. Free edges of leaflets of the tricuspid valve **20** are connected via the right ventricular chordae tendinae **34** to the right ventricular papillary muscles **36** in the right ventricle **14** for controlling the movements of the tricuspid valve **20**.

(20) After leaving the lungs, the oxygenated blood flows through the pulmonary veins **38** and enters the left atrium **16** of the heart **10**. The mitral valve **24** controls blood flow between the left atrium **16** and the left ventricle **18**. The mitral valve **24** is closed during ventricular systole when blood is ejected from the left ventricle **18** into the aorta **40**. Thereafter, the mitral valve **24** is opened to refill the left ventricle **18** with blood from the left atrium **16**. Free edges of leaflets **42***a*, **42***p* of the mitral valve **24** are connected via the left ventricular chordae tendinae **44** to the left ventricular papillary muscles **46** in the left ventricle **18** for controlling the mitral valve **30**. Blood from the left ventricle **18** is pumped through the aortic valve **26** into the aorta **40**, which branches into arteries leading to all parts of the body except the lungs. The aortic valve **26** includes three leaflets **48** which open and close to control the flow of blood into the aorta **40** from the left ventricle **18** of the heart as it beats.

(21) FIGS. **2**A-**2**D depict a device **50** deployed in a native mitral valve **24** and mitral valve annulus **25** according to an embodiment of the invention. The device **50** has a support anchor **52** and a sealing element **54**. The support anchor **52** may be configured to permit free movement of the

native valve leaflets **42***a*, **42***p*. In FIG. **2**A, the native mitral valve **24** is in early diastole, with the native leaflets **42***a*, **42***p* starting to open and permit blood to flow from the left atrium **16** to fill the left ventricle **18**. In full diastole as depicted in FIG. **2**B, the mitral valve leaflets **42***a*, **42***p* are fully open. In early systole as depicted in FIG. **2**C, initial backward flow into the left atrium **16** pushes the mitral valve leaflets **42***a*, **42***p* back (upwards) and into engagement with each other and with the sealing element **54**. In full systole as depicted in FIG. **2**D, the native mitral valve leaflets **42***a*, **42***p* wrap tightly against the sealing element **54** under peak systolic pressure, effectively eliminating any regurgitation flow.

(22) A sealing element **60** according to an embodiment of the invention is depicted in FIGS. **3**A-**3**D and **3**E-**3**H. The sealing element **60** has a fully compressed state, depicted in FIGS. **3**A and **3**E, which may be useful for delivery via a catheter and where the sealing element **60** may be substantially tubular and have an initial/delivery length **62***a* and an initial/delivery maximum diameter **64***a*. The sealing element **60** can be expanded by shortening the length **62***b*, **62***c*, thus forcing the sealing element maximum diameter **64***a*, **64***b*, to increase, as depicted in FIGS. **3**B-**3**C and **3**F-**3**G. The sealing element **60** can reach its deployed length **62***d* and deployed maximum diameter **64***d*, depicted in FIGS. **3**D and **3**H. A sealing element **60** according to the invention may have an initial/delivery length **62***a* of between 15-60 mm; 40-50 mm; 30-55 mm; etc. The sealing element **60** may have an initial/delivery maximum diameter **64***a* of between 0-10 mm; 5-8 mm; 6 mm; etc. The sealing element **60** according to the invention may have a deployed length **62***a* of between 20-45 mm; 25-35 mm; around 30 mm; etc., and a deployed maximum diameter **64***a* of between 5-40 mm; 10-30 mm; 15-25 mm; etc. The ratio of height/length to diameter may vary, depending on the particular application. Example ranges for height/length to diameter ratios are: 10:1, or 5:1, or 3:1, and ranges therebetween, during delivery; and 1:4, or 1:2, or 1:1, or 2:1, and ranges therebetween, at deployment. Note that each of these ratios and ranges for a particular sealing element dimension (e.g., delivery diameter, deployed diameter, delivery height/length, deployed height/length, etc.) may be combined with any and all of the other sealing element dimensions in accordance with the invention.

(23) A sealing element and/or anchor element according to the invention may include radiopaque or other visualization markers to enhance user visualization of the device. For example, in the embodiment of FIGS. **3**A-**3**F, radiopaque or other visualization markers **63**, **65**, are included. Markers **63** are positioned toward the ends of the sealing element, and can be used to visualize the height/length of the sealing element. Perimeter markers **65** are positioned toward the perimeter of the sealing element toward its widest point, and can be used to visualize the diameter of the sealing element. All markers can also be used to monitor the position of the sealing element.

(24) The sealing element **60** may have an expansion control element **66**, the length of which can be adjusted to thereby control the overall length and thereby the expansion of the sealing element. The sealing element **60** may preferably have a sealing surface (not shown), which may be in the form of an outer covering that prevents the passage of blood therethrough. The sealing surface may be supported by an expandable support frame (not shown). Note that in the embodiment depicted in FIGS. **3**A-**3**D, the expansion control element **66** corresponds to the central axis of the sealing element **60**, about which the sealing element **60** may be axisymmetric during deployment and after expansion.

(25) An expandable support frame **70** in a so-called “slotted tube” configuration according to an embodiment of the invention is depicted in FIGS. **4**A and **4**B. As shown in FIG. **4**A, in its compressed/delivery configuration the support frame **70** comprises a substantially cylindrical body **72**, and a delivery length **74***a* and delivery maximum diameter **76***a*. The support frame **70** construction, with slits **78** between rib-like elements **80**, permits the support frame **70** to be radially expanded, with the rib-like elements **80** bending outward, with the support frame **70** reaching a deployed length **74***b* and deployed maximum diameter **76***b* as depicted in FIG. **4**B. Note that the deployed length **74***b* is less than the delivery length **74***a*, while the deployed maximum diameter

76*b* is greater than the delivery maximum diameter **76***a*. The delivery configuration is conducive to delivery via a catheter into a human heart. The addition of a sealing surface (not shown), such as in the form of an outer covering, to the support frame **70** will create a sealing element according to an embodiment of the invention. Expansion of the support frame **70** may be accomplished via mechanisms such as tethers, ratchets, etc., and/or may involve constructing the support frame **70** from shape-memory materials such as nitinol.

(26) A support frame **90** according to an embodiment of the invention may include a braided mesh sleeve, depicted in FIGS. **5**A-**5**C, where strands **92** (such as wires or other strand-like elements) are braided or otherwise formed into a mesh structure defining a generally tubular structure **94**. The mesh structure may cause the frame **90** to radially expand when its length is compressed, and to radially reduce when its length is reduced (in similar fashion to contractible finger traps used as children's toys). The support frame **90** has a distal end **96** and proximal end **98**, with a length **100***a*, **100***b*, **100***c* and maximum diameter **102***a*, **102***b*, **102***c*. The diameters **104**, **106** of the distal end **96** and proximal end **98** may be generally fixed, so that this portion of the support frame **90** is prevented from appreciably radially expanding even as the maximum diameter expands. As the support frame **90** is reduced in length from its delivery length **100***a* to its deployed length **100***c*, the support frame **90** increases in maximum diameter from its delivery maximum diameter **102***a* to its deployed maximum diameter **102***c*. Note that, depending on how the support frame is formed (e.g., how the strands **92** are braided, the materials used, etc.), the central portion **108** of the support frame **90** may retain a substantially tubular shape even as the central portion **108** and support frame **90** are radially expanded. In other embodiments, the support frame **90** may form a more spherical shape, of even shapes having varying radial dimensions around a circumference of the frame **90**. The addition of a sealing surface (not shown), such as in the form of an outer covering, to the support frame **90** will create a sealing element according to an embodiment of the invention. Expansion of the support frame **90** may be accomplished via mechanisms such as tethers, ratchets, etc., and/or may involve constructing the support frame **00** from shape-memory materials such as nitinol.

(27) Support frames and sealing elements according to the invention may be configured to form non-circular profiles when viewed from the top. For example, in the embodiment of FIGS. **6**A-**6**D, a sealing element **110** which may have side profiles during delivery and expansion similar to that depicted in FIGS. **3**A-**3**D, but instead of having the circular top profiles as depicted in FIGS. **3**E-**3**H instead forms an elongated/elliptical top profile when deployed, where a major diameter **112***d* is much larger than the minor diameter **114***d* when the sealing element **110** is fully expanded as depicted in FIG. **6**D. Such a non-circular/elliptical sealing element may include radiopaque or other visualization markers **115**, **117**, e.g., positioned toward the sealing element perimeter and along/adjacent the major axis **115** and/or minor axis **117**, via which a user can visualize not just the dimensions (major diameter and/or minor diameter) but also the rotational orientation of the sealing element about its central axis and with respect to the anchor element and/or native valve. During delivery, the maximum diameter may be between 0-10 mm; 5-8 mm; 6-7 mm; etc. The expanded minor diameter **114***d* may be between 5-20 mm; 10-15 mm; etc., and the expanded major diameter **112***d* may be between 10-40 mm; 15-40 mm; 20-30 mm; etc.

(28) Expansion of sealing elements may be controlled via the use of various mechanisms. For example, as depicted in FIG. **7**A, an internal tethering mechanism **120** may comprise a tether line **122** with a first end **124** that may be secured to an end of a support frame (not shown) and a second end **126** that may passed through an opposing end of the support frame. By pulling on the second end **126** and thereby shortening the length of the internal tether that lies within the support frame, the support frame is reduced in length and radially expanded. The internal tethering mechanism may include a lock, such as a knot, one-way lock, or other locking mechanism (not shown), at the opposing end of the support frame for securing the internal tether at the desired length that corresponds to the desired shape (i.e., maximum diameter/length) of the deployed sealing element.

Such a tether may be used with any and all sealing elements and support frames of the invention, and may be particularly useful for sealing elements having self-expanding support frames where the tether acts to restrain and control expansion of the support frame.
(29) Expansion of a sealing element according to the invention may be controlled by a generally rigid rod **130** that resists both tension and compression, as depicted in FIG. **7**B. A first end **132** of the rod **130** may be secured to a first end of a support frame (not shown), with the rod **130** passing through a locking mechanism **134** at a second end of the support frame with the second end **134** of the rod **130** on the far side of the locking mechanism from the first end **132**. The rod **130** may include tooth-like elements **136** that are selectively engaged by the locking mechanism **134**. For example, the locking mechanism **134** may permit the tooth-like elements to pass outwardly through the locking mechanism (as may be desired to shorten the rod **130** and thereby shorten and radially expand a support frame), but prevent the tooth-like elements from passing inwardly through the locking mechanism.
(30) Another option for controlling expansion of sealing element is a linear screw mechanism **140**, where an elongated screw **142** has a first end **144** secured to a first end of a support frame (not shown). A screw-receiving nut **146** is secured via extenders **148** to a second end of the support frame. Either the screw-receiving nut **146** or the screw first end **144** is rotatably secured to the support frame, permitting rotation of the screw with respect to the screw-receiving nut which thereby advances/retracts the screw **142** through the screw-receiving nut **146** and thereby adjusts the length of the linear screw mechanism and the length/expansion of the support frame.
(31) FIG. **8**A depicts a perspective view of a device **150** that is configured for deployment with portions secured in the atrium, ventricle, and valve annulus of a patient. An anchor element **152** anchors and supports a sealing element **154**. The anchor element **152** has an atrial portion **156** with atrial arms **158** terminating in atrial arm distal ends **160**, with the atrial arms **158** sized and shaped to extend across the lower portion of the atrium and engage the valvular annulus from the atrial side. A ventricular portion **162** has ventricular arms **164** terminating in ventricular arm distal ends **166**. The ventricular arms **164** may be sized and shaped to extend around the “swing” area through which the native valve leaflets swing as the heart beats, so as not to engage against the native valve leaflets as the valve leaflets open and shut. The ventricular arms **164** curve around the leaflets swing area and then curve back upward to engage the valvular annulus from the ventricular side. Note that the anchor element **152** depicted when expanded in air or otherwise expanded in the absence of compressive forces (such as compressive forces that might be present when deployed in a heart) is axisymmetrical when viewed directly from above or below, as seen in FIGS. **8**B-**8**C, so that during deployment the surgeon or other user does not have to worry about whether the anchor element **152** of the device **150** is rotated around its central axis to a proper deployment position with respect to the native valve and valve annulus. In the particular embodiment depicted, the expanded diameter **161** of the upper anchor element **156** is slightly larger than the expanded diameter **167** of the lower anchor element **162**, although the invention is not limited to this configuration. The lower anchor expanded diameter **167** may be between 25 and 60 mm, and the upper anchor expanded diameter **161** may be between 30 and 70 mm. Note that when expanded in an actual heart, the individual arms **158**, **164** of the upper and lower anchor elements **156**, **162** will be compressed or otherwise distorted by their interaction with the heart tissue, so that each arm may be contorted by the heart tissue into a shape slightly different from the shapes of other arms.
(32) The anchor element **152** of FIG. **8**A includes a central portion **168** which passes through the sealing element **154**. The central portion **168** may be configured to act as a support frame for the sealing element, and/or may be configured to be shortened (e.g., by incorporating shortening mechanisms such as those in FIGS. **7**A-**7**C) in order to effectuate shortening and thereby expansion of the sealing element **154**. In the particular embodiment depicted, the sealing element central portion **168** surrounds an axis of the sealing element **154** and of the anchor element **152**. Note that the sealing element **154** in top profile (i.e., about its central axis) may be axisymmetric (e.g.,

circular) as in FIGS. **3**E-H, or may be non-circular (e.g., elliptical) as in FIGS. **6**A-**6**D. The sealing element **154**, especially if it is non-circular/elliptical in top profile, may be configured for axial rotation (selective or responsive to body operation such as blood flow/valve operation) with respect to the anchor element **152**, and a lock may be provided that can be activated to prevent further rotation of the sealing element **154** with respect to the anchor element **152**. With such rotation, the surgeon or other user could deploy the axisymmetric anchor element and confirm the anchor element's proper deployment (via direct or indirect viewing, e.g., radioscopy), and then rotate the sealing element to the desired rotational position, and then lock the sealing element at that desired position.

(33) FIGS. **8**D-**8**G depict side views of left atrium **16**, left ventricle **18**, mitral valve **24**, and mitral valve annulus **25**, with the device **150** at various stages of deployment therein according to an embodiment of the invention. The device **150** is secured within a distal end **170** of a delivery catheter **172**, and the distal end **170** is advanced percutaneously or minimally-invasively into the mitral valve **24** of the patient via the left atrium **16** of the heart **10**, as in FIG. **8**D. In FIG. **8**E, the ventricular anchor portion **166** is released from the catheter **172** and expanded into contact with heart tissue below the mitral valve **24**. In FIG. **8**F the sealing element **154** and central anchor portion **168** are positioned between the leaflets **42***a*, **42***p* of the mitral valve **24**. At this point the sealing element **154** may be expanded, or expansion may not occur until after deployment of the atrial anchor portion **156**. Although in FIG. **8**F the sealing element is depicted as expanded, note that expansion thereof may occur after deployment of the atrial anchor portion. In FIG. **8**G, atrial anchor portion **156** is expanded into contact with heart tissue above the mitral valve **24** as the device **150** is fully released from the catheter **172**.

(34) With the position and function of the device **150** confirmed (such as via radioscopy and/or other methods of remote viewing), the catheter **172** is then withdrawn from the patient. Although a percutaneous delivery via the atrial side is depicted, note that other deployment approaches are also within the scope of the invention, including transapical approaches.

(35) FIG. **9**A depicts a device **180** that is configured for deployment with anchor portions secured entirely in the atrium of a patient. An anchor element **182** anchors and supports a sealing element **184**. The anchor element **182** has an upper atrial portion **186** with upper atrial arms **188** terminating in upper atrial arm distal ends **190**. The upper atrial arms **188** are sized and configured to curve upward through the center of the atrium and outward to engage an upper surface of the atrium. The upper atrial arms **188** may include intermediate curves **189** that can act as shock absorbers, flexing and bending to permit the anchor element **182** to be compressed responsive to atrial shape changes as the heart beats. A lower atrial portion **192** has lower atrial arms **194** terminating in lower atrial arm distal ends **196**. The lower atrial arms **194** may be sized and shaped to extend outward and across the lower portion of the atrium and engage the valvular annulus from the atrial side. Note that the anchor element **182** depicted may be axisymmetric when expanded in the absence of compressive or other distorting forces, so that during deployment the surgeon or other user does not have to worry about whether the anchor element **182** of the device **180** is rotated around its central axis to a proper deployment position with respect to the native valve and valve annulus. For example, when expanded without compressive/distorting forces (e.g., expanded in air), the anchor element **182** is symmetrical when viewed from above or below, as depicted in FIGS. **9**B-**9**C. In the particular embodiment depicted, the expanded diameter **191** of the upper anchor element **186** is generally equal to the expanded diameter **197** of the lower anchor element **192**, although other diameters are also within the scope of the invention. The upper anchor expanded diameter **191** and lower anchor expanded diameter may be in the range between 30 and 70 mm. Note that when expanded in an actual heart, the individual arms **188**, **194** of the upper and lower anchor elements **186**, **192** will be compressed or otherwise distorted by their interaction with the heart tissue, so that each arm may be contorted by the heart tissue into a shape slightly different from the shapes of other arms.

(36) The anchor element **182** of FIG. **9**A includes a sealing element support portion **200** which passes below the atrial portions **186**, **192** and through the sealing element **184**, and is configured to hold the sealing element **184** at a desired position within the native valve annulus after the other anchor portions **186**, **192** have been deployed in the desired heart chamber, such as the left atrium **16** depicted. In the particular embodiment depicted, the sealing element support portion **200** corresponds to an axis of the sealing element **184** and an axis of the anchor element **182**. The sealing element support portion **200** may be configured to act as a support frame for the sealing element **184**, and/or may be configured to be shortened (e.g., by incorporating shortening mechanisms such as those in FIGS. **7**A-**7**C) in order to effectuate shortening and thereby expansion of the sealing element **184**. Note that the sealing element **184** in top profile (i.e., about its central axis) may be axisymmetric (e.g., circular) as in FIGS. **3**E-H, or may be non-circular (e.g., elliptical) as in FIGS. **6**A-**6**D. The sealing element **184**, especially if it is non-circular/elliptical in top profile, may be configured for axial rotation (selective or responsive to body operation such as blood flow/valve operation) with respect to the anchor element **182**, and a lock may be provided that can be activated to prevent further rotation of the sealing element **184** with respect to the anchor element **182**. With such rotation, the surgeon or other user could deploy the axisymmetric anchor element and confirm the anchor element's proper deployment (via direct or indirect viewing, e.g., radioscopy), and then rotate the sealing element to the desired rotational position, and then lock the sealing element at that desired position.

(37) As depicted in FIGS. **9**D-**9**G, the device **180** may be deployed via a transapical approach into the heart **10**. The device **180** is secured within a distal end **204** of a delivery catheter **202**, and the catheter distal end **204** is advanced via an opening **206** in the heart wall **11** (e.g., via an opening in a wall of the left ventricle **18**, such as a transapical opening at the apex) and into the mitral valve **24** of the patient. In FIG. **9**E, the upper atrial anchor portion **186** is expanded into contact with heart tissue toward the upper area of the left atrium **16** as the device **180** begins to be released from the catheter **202**. In FIG. **9**F the lower atrial anchor portion **192** is released from the delivery catheter **202** and expanded into contact with the lower portion of the atrium **16**. FIG. **9**G depicts the sealing element **184** and sealing element support portion **200** positioned between the leaflets of the mitral valve **24**, and the sealing element **184** is expanded therebetween. The device **180** is fully released from the catheter **202**. With the position and function of the device **180** confirmed (such as via radioscopy and/or other methods of remote viewing), the catheter **202** may then be withdrawn from the patient and the opening **206** closed via suture **208** and/or other methods/devices for closing openings.

(38) FIG. **10**A depicts a device **210** having a sealing element **212** anchored via a tether **214** and anchor **216**. The device **210** may be positioned in the distal end **218** of a delivery catheter **220** and advanced into the heart. In one embodiment, the device **210** is advanced transapically into the left ventricle **18** and to the mitral valve **24** via an opening **222** in the heart wall, where the opening **222** may be at the heart apex **19**, as depicted in FIG. **10**B. The sealing element **212** may be released from the catheter **216** at a position between the native valve leaflets **42***a*, **42***p*, as depicted in FIG. **10**C. FIG. **10**D depicts the catheter **220** withdrawn as the tether **214** is released from the catheter distal end **218**, with the tether **214** being rigid or flexible depending on the desired application. As the catheter is removed from the heart **10**, the anchor **216** is deployed in the heart wall **11**, as shown in FIG. **10**E. The user can then adjust the length of the tether **214** before locking the tether **214** to the anchor **216** at the desired length. Note that locking of the tether **214** to the anchor at the desired tether length can be achieved by known methods, such as via sliding locking mechanisms that initially allow the tether to slide through the lock but then prevent further movement, via knots in the tether line, etc.

(39) Note that the sealing element **212** may be selectively expanded, fully or partially, at various times during deployment, such as when initially released from the delivery catheter; after the tether is extended; after the anchor is secured; or after the tether length is finalized. The tether length can

also be adjusted at various times. For example, the sealing element **212** may be partially expanded when released from the catheter, with full expansion occurring just prior to or as the tether length is finalized in order to confirm proper sealing of the valve leaflets against the sealing element via radioscopy, etc. Although a transapical delivery via the ventricle and heart apex is depicted, note that other deployment approaches are also within the scope of the invention, including percutaneous approaches via blood vessels and/or the atrium side.

(40) An anchor frame **230** according to the invention may be formed of a memory material such as nitinol, and/or may have a central portion **232** configured to serve as a support frame for a sealing element. For example, as depicted in FIG. **11**A, a frame **230** is formed as a generally tubular form of shape memory material with a central portion **232**, upper portion **234**, and lower portion **236**, with each portion composed of slits **238***a-c* between rib-like elements **240***a-c*. The central portion **232** may be radially expanded into a desired shape by bending the rib-like elements **240***b* outward as depicted in FIG. **11**B, and the desired shape can be set in the memory material via known methods. The upper and lower portions **234**, **236** may similarly be formed to the desired shapes as in FIG. **11**C, and the desired shapes set to the memory material. Note that the rib-like elements **240***a-c* are configured to extend out to form arms and other frame/anchor structures. The addition of a fluid-blocking covering **242** to the frame central portion **232** creates the sealing element **244**. Note that the device may have a top opening **246** and/or a lower opening **248** in the sealing element portion, which permits some blood to flow into and out of the sealing element but avoids pooling of blood within it. Alternatively, one or both of the top and bottom of the sealing element portion may be entirely closed.

(41) Anchor frames according to the invention may be formed of various biocompatible materials, including metals and polymers. For example, memory materials such as Nitinol may be used, thereby forming an anchor frame that can be compressed onto/into a catheter for minimally-invasive/percutaneous delivery and then will expand to its “memorized” shape upon release from the catheter. Non-memory materials such as stainless steel or cobalt chromium are also within the scope of the invention. The anchor frame may include a biocompatible covering, such as of a Dacron or other fabric. The biocompatible covering may encourage tissue ingrowth to promote tissue anchoring. The biocompatible covering may alternatively resist tissue ingrowth.

(42) Sealing elements according to the invention may be formed of various biologically compatible materials, including metals, fabrics, plastics, and tissue. Some materials that may be used for such sealing elements include materials currently used in forming leaflets of prosthetic heart valves. For example, synthetic materials (e.g., polymers such as thermoplastic elastomers or resins, including polyurethane and silicone, etc.), natural/treated tissue (e.g., valve leaflet tissue, bovine or equine pericardium, etc.), fabrics (e.g., Dacron), etc. may be used.

(43) Sealing elements may preferably wrap around the exterior and/or interior of any anchor portions that serve to directly support or frame the sealing element, so that any wireform/rib-like elements of the support frame portion are covered and/or encapsulated by the sealing element material in order to prevent the native valve leaflets from contacting any frame elements of the sealing element support frame, etc.

(44) During deployment of a device according to the invention, such as the deployment procedures depicted in FIGS. **8**D-**8**G or **9**D-**9**G or **10**B-**10**E, the surgeon or other user may controllably expand the sealing element to a desired diameter/length, or the sealing element may be configured to self-expand to a pre-set configuration such as upon release from the delivery catheter. For example, if the sealing element comprises a memory-material (e.g., nitinol) support frame, that support frame may automatically expand to a pre-set memory material configuration (length/diameter) upon release from radial constraints such as a delivery lumen of the delivery catheter. Alternatively, the surgeon or other user may actively control the expansion of the sealing element to a specifically selected diameter/length, such as by activating length/diameter adjusting elements such as those depicted in FIGS. **7**A-**7**C. Before, during, and/or after sealing element expansion, the surgeon/user

may actively monitor heart valve function and/or leaflet coaptation (with the sealing element) and/or sealing element position/size via various techniques, such as fluoroscopy or other visualization techniques (including the use of radiopaque markers on the sealing element). The surgeon/user may select the final diameter/length of the sealing element based on the information on heart valve function and/or leaflet coaptation. The surgeon/user may also selectively rotate the sealing element about its central axis with respect to the support anchor (before, during, or after deployment of the support anchor) in order to better position the sealing element between the native valve leaflets, such as where the sealing element has an elliptical top profile as depicted in FIGS. **6**A-**6**D. For example, with such an elliptical profiled sealing element, the user may deploy the anchor element, then rotate the sealing element (expanded or unexpanded) with respect to the anchor element and native valve in order to position the minor diameter to extend generally perpendicular to the anterior and posterior mitral valve leaflets, and to position the major diameter to extend between the valve commissural points. Once the desired rotational position is achieved, the sealing element can be locked in that position.
(45) If the user (e.g., surgeon or other medical staff) is not satisfied with the initial positioning of all or part of the device, the device or parts thereof may be withdrawn (completely or partially) into the catheter and then re-deployed at the desired position. For example, if after initial deployment the sealing element is positioned too high or too low with respect to the mitral valve leaflets, the device or parts thereof can be at least partially withdrawn into the catheter and then re-deployed at a position higher or lower than the previous position. Similarly, if the user is not satisfied with the deployed size of the sealing element, he/she can adjust the length/radius of the sealing element until the desired sealing/coaptation with the native leaflet is achieved. Also, for sealing elements that are non-circular/elliptical in top profile, the user can modify the rotational position of the sealing element to a desired rotational position that may optimize valve function.
(46) Radiopaque markers or other visibility-enhancing markers may be included with the device in order to make the device and key elements thereof more clearly visible when the device is deployed or inspected using fluoroscopy or other visualization techniques. For example, enhanced visibility markers such as radiopaque markers may be secured to portions of the sealing element and/or the anchor elements, etc.
(47) Note that FIGS. **3**B-**3**D, **8**E, **9**A, and **11**A-**11**D are computer-generated to-scale drawings where dimensions are to scale within each drawing. All dimensions listed are by way of example, and devices according to the invention may have dimensions outside those specific values and ranges. Although in some of the drawings the sealing element material is depicted as extending between frame portions but with the frame portions uncovered, in various embodiments of the invention (including each of the embodiments of the invention depicted in the drawings) the sealing element material may preferably wrap around the exterior and/or interior of the support frame portions, so that the wireform/rib-like frame elements are covered and/or encapsulated by sealing element material in order to prevent the native valve leaflets from contacting any frame element.
(48) Although the specific embodiments discussed above are directed toward mitral valve repair, the invention may also be applicable for use in repairing other heart valves, including the aortic, tricuspid, and pulmonary valves.
(49) Unless otherwise noted, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. In order to facilitate review of the various embodiments of the disclosure, the following explanation of terms is provided:
(50) The singular terms “a”, “an”, and “the” include plural referents unless context clearly indicates otherwise. The term “or” refers to a single element of stated alternative elements or a combination of two or more elements, unless context clearly indicates otherwise.
(51) The term “includes” means “comprises.” For example, a device that includes or comprises A and B contains A and B, but may optionally contain C or other components other than A and B.

Moreover, a device that includes or comprises A or B may contain A or B or A and B, and optionally one or more other components, such as C.
(52) The term “subject” refers to both human and other animal subjects. In certain embodiments, the subject is a human or other mammal, such as a primate, cat, dog, cow, horse, rodent, sheep, goat, or pig. In a particular example, the subject is a human patient.
(53) Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present disclosure, suitable methods and materials are described below. In case of conflict, the present specification, including terms, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.
(54) In view of the many possible embodiments to which the principles of the disclosed invention may be applied, it should be recognized that the illustrated embodiments are only examples of the invention and should not be taken as limiting the scope of the invention. Rather, the scope of the invention is defined by the following claims. We therefore claim as our invention all that comes within the scope and spirit of these claims.

Claims

1. A sealing element configured for anchoring between native valve leaflets to improve valve function, the sealing element comprising: a mesh support frame comprising a delivery configuration and an expanded configuration, wherein in the delivery configuration the mesh support frame is substantially tubular with a delivery diameter, and wherein in the expanded configuration the mesh support frame is radially expanded to an expanded central diameter at a center portion thereof while end portions of the mesh support frame remain in the delivery diameter, wherein the expanded diameter is at least twice the delivery diameter; a covering over the mesh support frame, wherein the covering inhibits passage of blood therethrough; an anchor member comprising an atrial portion configured to engage an atrial side of the valve and a ventricular portion configured to engage a ventricular portion of the valve; and wherein the sealing element is configured to block flow therethrough in both diastole and systole.
2. The sealing element of claim 1, wherein the expanded central diameter is at least three times the delivery diameter.
3. The sealing element of claim 1, wherein the expanded central diameter is at least five times the delivery diameter.
4. The sealing element of claim 1, wherein the sealing element in the expanded configuration comprises an axisymmetrical top profile.
5. The sealing element of claim 1, wherein the sealing element in the expanded configuration comprises an elongated top profile.
6. The sealing element of claim 1, wherein the sealing element in the expanded configuration comprises an elliptical top profile.
7. The sealing element of claim 1 wherein the mesh support frame comprises a shape having varying radial dimensions around a circumference of the frame.